

Multi-criteria evaluation of symmetric tesla-like valve flow channel geometry on proton exchange membrane fuel cell performance using analytic hierarchy process- entropy weight method

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ABSTRACT

To investigate the influence and relative importance of geometric parameters on the performance of proton exchange membrane fuel cells (PEMFCs) with Tesla-like valve flow fields, and ultimately obtain the optimal combination of design parameters to improve the performance of PEMFC. This study employs the Analytic Hierarchy Process – Entropy Weight Method (AHP-EWM) model to investigate the influence of geometric parameters (loop Angle θ , channel height H , and channel width W) on the performance of PEMFC. A three-dimensional physical model of the symmetrical Tesla-like valve flow channel proton exchange membrane fuel cells (STSF) PEMFC is established, and its performance is evaluated through three indicators: current density (I), pressure drop (ΔP), and oxygen uniformity index (UI) at the gas diffusion layer (GDL)/catalytic layer (CL) interface. Through numerical simulation and sensitivity analysis, combined with AHP (Subjective Weight) and EWM (Objective Weight), the comprehensive weights of each parameter and performance index are determined. The results show that the order of importance of each performance index is: I (0.58) > UI (0.22) > ΔP (0.2), the geometric parameters are W (0.44) > H (0.375) > θ (0.185). Sensitivity analysis indicates that W and H are more sensitive to ΔP and UI than θ . The final results indicate that the parameter combination of $\theta = 30^\circ$, $H = 1.2$ mm, and $W = 0.6$ mm does not yield the lowest pressure drop among all configurations. However, it achieves the best overall balance among current density, pressure drop, and oxygen distribution uniformity.

Glossary

Nomenclatures		Greek letters	
Abbreviation	Description	Abbreviation	Description
t	time [s]	ρ	density [kg/m^3]
$\vec{u}$	fluid velocity vector [m/s]	μ	dynamic viscosity coefficient [$\text{kg}/(\text{m}\cdot\text{s})$]
F	faraday's constant [C/mol]	δ	the sum of the activation overpotential and the overconcentration [V]
M_x	molecular weight of species x [kg/mol]	ϕ	electric potential [V]
i	volumetric transfer current density [A/m^2]	η	overpotential [V]
K_x	permeability of x [m^2]	ε	porosity

(continued on next column)

Glossary (continued)

Nomenclatures		Greek letters	
Abbreviation	Description	Abbreviation	Description
k^{eff}	effective thermal conductivity [$\text{W}/(\text{m}\cdot\text{K})$]	ζ	specific active surface area
I	current density [A/cm^2]	γ	concentration dependence
R_{or}	ohmic resistivity of media [$\Omega\cdot\text{m}$]	α	transfer coefficient
$P_{\text{H}_2\text{O}}$	gaseous water generation rate [kg/s]	σ	electrical conductivity or surface tension
h_{re}	reaction enthalpy change [kJ/mol]	ξ	stoichiometry ratio
r_w	rate of aqueous phase change [mol/s]	Subscripts	

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Glossary (continued)

Nomenclatures		Greek letters	
Abbreviation	Description	Abbreviation	Description
h_l	phase transition enthalpy [kJ/mol]	Abbreviation	Description
c_h	concentration of the species h [l/s]	M	momentum
D_h^{eff}	the effective diffusion coefficient of the species h [m ² /s]	AHP	Analytic Hierarchy Process
S_{hp}	concentration source term of species h [kg/(m ³ •s)]	EWM	Entropy Weight Method
j_a	anode volumetric exchange current density [A/m ³]	UI	oxygen uniformity index
j_c	cathode volumetric exchange current density [A/m ³]	a	anode side
R_{sp}	the source term of the electron current [A]	c	cathode side
R_{mp}	the source term of the proton current [A]	PEMFC	Proton exchange membrane fuel cells
C_x	concentration of species x [mol/m ³]	com	comprehensive
p	pressure [Pa]	ref	reference
ω_x	the mass fractions of x	H ₂	hydrogen
S_x	source term of x [kg/(m ³ •s)]	H ₂ O	water
c_p	specific heat capacity at constant pressure [J/(kg•K)]	O ₂	oxygen
$j_{a,c}$	the exchange current density between the anode and the cathode [A/m ²]		
j^{ref}	the reference exchange current density per active surface area [A/m ²]		

1. Introduction

With the rapid expansion of industrial production, fossil fuels such as petroleum and natural gas face issues of environmental pollution and resource depletion during their use in energy conversion systems. Therefore, developing a pollution-free and highly efficient energy utilization method has become an urgent objective for most countries today. Hydrogen energy is regarded as an effective energy carrier, and its technology represents an important approach to energy development [1]. Proton exchange membrane fuel cells (PEMFCs), which can efficiently convert chemical energy from fuels into electricity without causing pollution, are regarded as one of the most promising and practically valuable power sources [2,3]. However, numerous challenges must be addressed in the application of PEMFCs, including battery performance, fuel utilization efficiency, and water management [4].

The bipolar plate is one of the crucial components in PEMFCs, which not only removes excess water produced during reactions and collects electric current, but also distributes reactant gases and conducts heat [5]. Therefore, a more optimized flow field structure within the bipolar plate results in superior fuel cell system performance. Improving the geometric structure and shape of the flow field has become an essential aspect of enhancing PEMFC performance [6].

In the past, many researchers have conducted in-depth studies on flow field structures to improve the overall performance of fuel cells and reduce system operating costs. Due to issues such as uneven distribution of reactant gases and poor mass transfer capability in traditional flow field designs, flow field modifications incorporating baffles have been proposed to address these problems [7–9]. The results show that PEMFCs with baffles in the flow channels exhibit significantly improved performance, primarily because the induced secondary flow enhances

mass transfer toward the catalyst layer. Some researchers, inspired by biological structures in nature, have designed various types of leaf-vein-inspired bionic flow fields to overcome the limitations of traditional configurations [10–12]. Findings indicate that these bionic flow fields improve current density distribution, enable effective water management, and reduce gas flow resistance. Huang et al. [13] investigated the impact of an artery-inspired bionic flow field on PEMFC performance through simulation and experimentation. Compared to the conventional serpentine flow field, droplet accumulation at the corners was alleviated, and the droplet removal rate increased by 36.3%. Under actual operating temperatures, the pressure drop was reduced by 30.03% compared to the conventional serpentine flow field. Other researchers have proposed special structural designs for flow channels, such as Tesla valve-based flow fields. Aisyah Sarjuni et al. [14] designed a Tesla valve (TV) flow field and compared its performance with that of Modified Serpentine (MS), Modified Fishbone (MF), and Conventional Parallel (CP) flow fields. The results showed that at an inlet/outlet angle of 135°, the TV flow field demonstrated reduced pressure drop, more uniform reactant gas distribution, and enhanced water removal capacity. To further enhance PEMFC performance, Gong et al. [15] developed a multi-stage Tesla valve flow field (MSTV). The study revealed that the MSTV design achieved a higher average oxygen mass fraction at the cathode gas diffusion layers/ catalyst layers (GDL/CL) interface than the conventional parallel flow field (CPFF), and approached the performance of the conventional serpentine flow field (CSFF), while maintaining a lower pressure drop. Guo et al. [16] emphasized the importance of Z-direction mass transfer in Tesla valve flow fields. They conducted the first quantitative analysis of the effect of Z-direction velocity on mass transfer, demonstrating that enhanced flow perpendicular to the channels significantly improves oxygen supply to the catalyst layer and water management. Zhang et al. [17] proposed a large-scale dual-symmetric Tesla valve cathode flow field (DSTVFF) and optimized a series of structural parameters, including module arrangement, interior angles, and quantity. The results confirmed that the fourth-level dual-symmetric Tesla configuration (DSTVC) achieved a lower pressure drop and more uniform oxygen concentration distribution. The power density with a 30° inner angle was 17.6% higher than that with a 15° inner angle.

The aforementioned researchers achieved promising results by introducing alternative flow field structures that altered the single transport pattern of reactant gases in conventional designs. In addition to structural redesigns, optimized geometric parameters of the flow channels are also critical factors influencing PEMFC performance. Bachman et al. [18] designed parallel flow fields with varying channel lengths and found experimentally that the length of the flow channels significantly affects fuel cell performance and stability. Zeng et al. [19] optimized the flow channel geometry using a genetic algorithm, with output power as the objective and the widths of the channel's top and bottom as variables. The results showed that the optimal design was a trapezoidal channel, which significantly increased current density and improved reactant distribution compared to square channels. Liu et al. [20] used inlet and outlet heights as variables and applied a multi-objective genetic algorithm to optimize the flow channel structure, resulting in enhanced mass transfer pressure. Manso et al. [21] investigated PEMFC performance using ten flow channels with different cross-sectional aspect ratios. The results indicated that flow fields with higher aspect ratios provided more uniform current distribution and lower temperature gradients. Chiu et al. [22] studied the effects of different flow channel widths, heights, and aspect ratios on fuel cell performance. Their findings showed that reducing the channel width or height could enhance the performance of both parallel and serpentine flow fields and reduce concentration polarization. As flow field design evolves, the application of various optimization techniques has made notable contributions to improving PEMFC performance. Priya et al. [23] provided a comprehensive overview of various metaheuristic algorithms for parameter estimation in fuel cells. These include genetic

algorithms, particle swarm optimization, and artificial neural networks [24–26]. Xiao et al. [27] combined a Kriging model with genetic and particle swarm optimization algorithms to analyze the effect of the angle between the main and secondary channels in a bionic flow field on fuel cell performance. Experimental results showed that the optimized bionic flow field outperformed traditional designs under high current densities. Wang et al. [28] integrated backpropagation neural networks with genetic algorithms to examine how flow channel height and width in bionic flow fields affect mass transfer. Their findings indicated that channel width had a greater influence on voltage and liquid water saturation, while channel height affected pressure drop and reactant concentration at the outlet.

In summary, both the flow field configuration and the geometric parameters of the channels have a significant impact on PEMFC performance. Among them, the Tesla-like valve flow field, with its passive unidirectional flow characteristics, can achieve fluid directional transmission without additional valves, effectively reducing dead zones in the flow channel and enhancing secondary flow, thereby improving the distribution of reactants and water management. The geometric parameters of the flow channel of the Tesla-like valve (such as the loop Angle θ , flow channel height H , and width W) have the characteristic of multiple adjustable dimensions. However, most existing studies have focused on single structural improvements (such as multi-level design and Z-direction mass transfer), and have not systematically explored the coupling influence of these core geometric parameters on multiple performance indicators (current density, pressure drops, oxygen uniformity). This study combines Analytic Hierarchy Process – Entropy Weight Method (AHP-EWM) with One-Factor-At-a-Time sensitivity analysis (OFAT) for the first time. It not only determines the parameter weights but also quantifies the sensitivity of the parameters to each performance index (current density, pressure drop, oxygen uniformity), solving the problem that the traditional AHP-EWM only focuses on weights and ignores the interaction of parameters. For the three-dimensional structure of the symmetrical Tesla valve flow channel proton exchange membrane fuel cells (STSFC), a multi-parameter collaborative optimization scheme of “loop Angle – height – width” is proposed, filling the research gap in the systematic optimization of geometric parameters of this type of flow field (see Table 1).

2. Model development

This chapter aims to explore the performance of symmetrical Tesla-like valve flow channel proton exchange membrane fuel cells (STSFC). It begins with an overview of the model construction framework and the modeling methods used in the research. The process starts with building a physical model, which clarifies the battery's geometric structure, key components, and operating environment. This provides a basis for the mathematical description and numerical simulation. It should be noted that in this study, Comsol 6.3 Multiphysics is adopted for the numerical simulation of PEMFC multiphysics field. The following section will detail the design of the physical model.

2.1. Physical model

Tesla valves are commonly used in passive fluid control systems, and

Table 1
Geometric Parameters of the Model.

Parameter	Value	Unit
loop angle (θ)	90	°
Channel length (L)	30	mm
Channel width (W)	0.6	mm
Channel height (H)	1	mm
A/CGDL thickness	0.55	mm
A/CCL thickness	0.05	mm
MEM thickness	0.1	mm

their unidirectional flow characteristics improve with greater structural symmetry [29]. In this study, the loop baffles in the Tesla valve structure are simplified to construct a three-dimensional physical model of a single-channel PEMFC with a symmetric Tesla-like valve structure. This model includes anode (a)/cathode (c) flow channels, anode/cathode gas diffusion layers (GDLs), anode/cathode catalyst layers (CLs) and proton exchange membranes (MEM), as illustrated in Fig. 1.

2.2. Mathematical model

After the construction of the physical model is completed, this chapter will quantitatively describe the multi-physics coupling process inside STSFC through a mathematical model to achieve numerical simulation and analysis of performance. This mathematical modeling process needs to follow the logical progression of “simplified hypothesis – core equation – boundary constraints”. The following will elaborate on the assumptions, control equations and boundary conditions of the mathematical model in sequence according to this logic.

2.2.1. Model assumptions

The computational modeling of PEMFCs is typically based on a set of assumptions that abstract the physical processes into mathematical equations. Accordingly, this study adopts several widely accepted assumptions, as detailed below [30–32]:

- (1) The gases involved in the electrochemical reactions are considered incompressible ideal gases, and the flow in the channels is assumed to be laminar.
- (2) The PEMFC operates under steady-state conditions.
- (3) The GDL and CL are treated as homogeneous and isotropic porous media.
- (4) The effect of gravity on the PEMFC is neglected.
- (5) Diffusion of reactant gases through the membrane is considered negligible.
- (6) The PEMFC operates under isothermal conditions.

2.2.2. Governing equations

Based on the above assumptions, the main governing equations are presented below, with detailed explanations of each term provided in Table 2.

Continuity equation [33]:

$$\frac{\partial(\epsilon\rho)}{\partial t} + \nabla \cdot (\epsilon\rho \vec{u}) = S_{c_p} \quad (1)$$

where, ϵ is the porosity, ρ is the fluid density, $\vec{u}$ is the fluid velocity vector, t is the time S_{c_p} is the mass source term. The specific explanations are shown in Table 3.

Momentum Conservation Equation [34]:

$$\frac{\partial(\epsilon\rho\vec{u})}{\partial t} + \nabla \cdot (\epsilon\rho\vec{u}\vec{u}) = -\epsilon\nabla p + \nabla \cdot (\epsilon\mu\nabla\vec{u}) + S_{M_p} \quad (2)$$

where, μ is the dynamic viscosity, S_{M_p} represents the momentum source term.

Energy conservation equation [35]:

$$\frac{\partial(\epsilon\rho c_p T)}{\partial t} + \nabla \cdot (\epsilon\rho c_p \vec{u} T) = \nabla \cdot (k^{eff} \nabla T) + S_{Q_p} \quad (3)$$

where, c_p is the specific heat capacity at constant pressure, T is the temperature, k^{eff} is the thermal conductivity, S_{Q_p} is the volumetric source term of energy.

Species conservation equation [31]:

$$\frac{\partial(\epsilon C_h)}{\partial t} + \nabla \cdot (\epsilon \vec{u} C_h) = \nabla \cdot (D_h^{eff} \nabla C_h) + S_{h_p} \quad (4)$$

where, S_{h_p} is the species source term, C_h is the component concentration

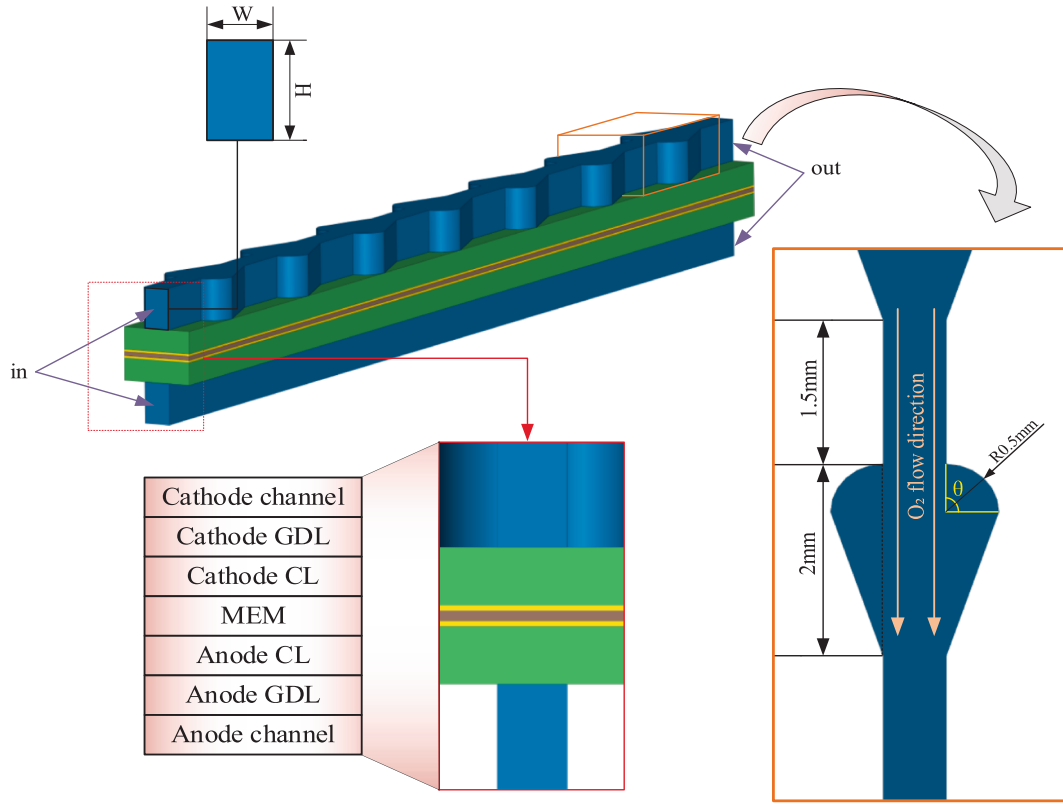


Fig. 1. Schematic diagram of a single-channel PEMFC device.

Table 2
Formulations of relevant parameters.

Parameters	Formulation
Bulter–Volmer	$R_{\text{wa}} = \left(\zeta_{\text{a,a}}^{\text{ref}} \right) \left(\frac{C_{\text{H}_2}}{C_{\text{ref}}} \right)^{\gamma_a} \left[e^{\frac{\alpha_a F \eta_a}{RT}} - e^{-\frac{\alpha_c F \eta_a}{RT}} \right]$ $R_{\text{oc}} = \left(\zeta_{\text{c,c}}^{\text{ref}} \right) \left(\frac{C_{\text{O}_2}}{C_{\text{ref}}} \right)^{\gamma_c} \left[-e^{-\frac{\alpha_a F \eta_c}{RT}} + e^{\frac{\alpha_c F \eta_c}{RT}} \right]$ $\eta_a = \phi_{\text{sp}} - \phi_{\text{mp}}$ $\eta_c = \phi_{\text{sp}} - \phi_{\text{mp}} - V_{\text{oc}}$
Phase change rate	$r_c = \begin{cases} (1-s) \frac{P_{\text{wv}} - P_{\text{sat}}}{RT} M_{\text{H}_2\text{O}} P_{\text{wv}} > P_{\text{sat}} \\ s \frac{P_{\text{wv}} - P_{\text{sat}}}{RT} M_{\text{H}_2\text{O}} P_{\text{wv}} < P_{\text{sat}} \end{cases}$
Capillary pressure	$p_c = \begin{cases} \frac{\sigma_t \cos \theta_c}{\left(\frac{K_a}{\varepsilon} \right)^{0.5}} [1.417(1-s) - 2.12(1-s)^2 + 1.263(1-s)^3] & \theta_c < 90^\circ \\ \frac{\sigma_t \cos \theta_c}{\left(\frac{K_a}{\varepsilon} \right)^{0.5}} (1.417s - 2.12s^2 + 1.263s^3) & \theta_c > 90^\circ \end{cases}$

Table 3
Source terms of the conservation equations.

Governing equations	Source terms	Equations
Continuity equation	S_{C_p}	$\begin{cases} 0 \text{ (GDLs, flowchannel)} \\ -\frac{M_{\text{H}_2}}{2F} i_a \text{ (anodeCL)} \\ \frac{M_{\text{H}_2\text{O}}}{2F} i_c - \frac{M_{\text{O}_2}}{4F} i_c \text{ (cathodeCL)} \end{cases}$
Momentum Conservation Equation	S_{M_p}	0 (flow channel) According to Darcy's law: $S_{M_p} = -\frac{k_p}{\mu} \nabla p$
Energy conservation equation	S_{Q_p}	$I_p^2 R_{\text{or}} + \beta P_{\text{H}_2\text{O}} h_{\text{re}} + r_w h_l + j_{a,c} \delta p$
Species conservation equation	S_{h_p}	$\begin{cases} 0 \text{ (GDLs, flowchannel)} \\ -\frac{M_{\text{H}_2}}{2F} j_a \text{ (anodeCL)} \\ \frac{M_{\text{H}_2\text{O}}}{2F} j_c - \frac{M_{\text{O}_2}}{4F} j_c \text{ (cathodeCL)} \end{cases}$
Electric charge conservation equation	$R_{\text{sp}}, R_{\text{mp}}$	$\begin{cases} -R_{\text{wa}} \text{ (for the solid phase)} \\ R_{\text{wa}} \text{ (for the membrane phase)} \end{cases} \text{ (anode)}$ $\begin{cases} R_{\text{oc}} \text{ (for the solid phase)} \\ -R_{\text{oc}} \text{ (for the membrane phase)} \end{cases} \text{ (cathode)}$

of species h, D_h^{eff} is the effective diffusion coefficient of the species h.

Electric charge conservation equation [36]:

$$\nabla \cdot (\tau_{\text{sp}} \nabla \phi_{\text{sp}}) + R_{\text{sp}} = 0 \quad (5)$$

$$\nabla \cdot (\tau_{\text{mp}} \nabla \phi_{\text{mp}}) + R_{\text{mp}} = 0 \quad (6)$$

where, τ_{sp} is the solid phase conductivity, τ_{mp} is the membrane phase conductivity, ϕ_{sp} is the solid phase potential, ϕ_{mp} is the membrane phase potential, R_{sp} is the electronic current source term, R_{mp} is proton current source term.

2.2.3. Boundary condition

In this study, boundary conditions are defined for the numerical simulation to solve the governing equations.

- (1) Assuming no-slip boundary at the channel walls.
- (2) The anode side potential reference is set to zero potential.
- (3) Symmetry boundary conditions are applied on both sides of the x-axis of the flow channel.
- (4) The cathode side voltage is 0.05 V as a step, and the gradient scan is performed in the range of 0.15 V–1 V.

The inlet and outlet of the flow channel are defined as mass flow inlet and pressure outlet, respectively. The corresponding values are determined according to the following equations [37]:

$$m_{a,c} = \frac{\rho_{a,c} \xi_{a,c} I_{ref} A_{act}}{2F\gamma_{H_2,O_2}} \quad (7)$$

$$C_{H_2} = \frac{P_a + \Delta P_a - RH_a P_{sat}}{RT} \quad (8)$$

$$C_{O_2} = 0.21 \frac{P_c + \Delta P_c - RH_c P_{sat}}{RT} \quad (9)$$

$$C_{H_2O} = \frac{RH_{a,c} P_{sat}}{RT} \quad (10)$$

$$\rho_{a,c} = \gamma_{H_2,O_2} M_{H_2,O_2} + C_{a,c}^{H_2O} M_{H_2O} \quad (11)$$

In addition, the specific model physical parameters and operating conditions applied in this study are provided in the following Table 4.

2.3. Model validation

To ensure that the established mathematical model can accurately reflect the actual operational characteristics of STSFC, model verification needs to be carried out from two core dimensions: “computational reliability” and “mesh independence”. The former verifies the authenticity of the simulation results by comparing them with experimental data, while the latter eliminates the interference of grid density on the calculation accuracy through grid independence analysis. Together, they constitute the key support for the validity of the model. First, through the simulation model verification stage, the core performance indicators of the simulation and the experiment will be compared to initially confirm the accuracy of the model. On this basis, further grid independence tests are carried out to determine the optimal grid scheme for subsequent simulation analysis.

2.3.1. Simulation model verification

Although CFD numerical simulation can effectively predict the state of PEMFC, experimental verification is crucial for ensuring the reliability of the numerical model. This study refers to the experimental data of Ubong et al. [38] and constructs the model under the same physical field conditions. Our simulation results are compared with the data reported by Zhi et al. [39] during the verification process, we analyze two experimental conditions with pressures of 1 atm and

2.7 atm. The comparison between the simulation results and the experimental data is shown in Fig. 2. The results show that the polarization curves of the four groups of data have consistent trends and good consistency, with a relative error of approximately 4.4 %. This indicates that the numerical simulation method meets the requirements of calculation accuracy and is suitable for subsequent simulation research.

2.3.2. Mesh independence test

During the simulation process, different numbers and qualities of grids will produce different results [40]. To ensure the accuracy of the numerical results and improve the computational efficiency, a grid independence test is conducted to determine the optimal grid density. The mesh quality distribution of the STSFC model is shown in Fig. 3 (a). In this study, the current density is tested under different grid numbers (87243, 160847, 311867, 460249) within the operating voltage range of 1.0 V to 0.45 V to verify the grid independence of cell performance. The test results are shown in Fig. 3 (b). It can be found that after 311,867 grids, the current density curve changes very little. The current density change rate of the 311,867 and 460,249 grid cells at each voltage is less than 1 %, indicating that both meet the grid encryption requirements. Moreover, the single simulation time of 311,867 grids is approximately 8 h, which saves 33 % of the computing cost compared to 460,249 grids (12 h). In order to enhance the efficiency and accuracy of the calculation and reduce the calculation time, 311,867 grid cells are selected as the final computing grid.

2.4. Methods (AHP-EWM multi-criteria decision framework)

To achieve multi-dimensional and scientific quantitative evaluation of the performance of STSFC, this study constructs an AHP-EWM combined weight multi-criteria decision-making framework. By combining subjective judgment with objective data, it avoids the limitations of a single weighting method and ensures the rationality of weight distribution and the reliability of evaluation results. The detailed procedure is illustrated in Fig. 4. The following will elaborate on the specific implementation steps of each method in sequence according to this process.

2.4.1. Analytical hierarchy process

The Analytic Hierarchy Process (AHP) is a method that decomposes complex multi-criteria decision-making problems into a hierarchical structure and determines weights by integrating subjective judgment with mathematical computation. It allows the decision problem to be broken down into a hierarchy consisting of the overall goal, various

Table 4
Properties and parameters.

Parameters	Values	Unit
Operation pressure	101×10^3	Pa
Operation temperature	453.15	K
Faraday's constant	96,487	C/mol
Stoichiometric ratio of the anode side	1.2	—
Stoichiometric ratio of the cathode side	2	—
H ₂ mass fraction in the anode	0.96268	—
O ₂ mass fraction in the cathode	0.20216	—
H ₂ O mass fraction in the cathode	0.037319	—
Open circuit voltage	1	V
Porosity of GDLs	0.4	—
Porosity of CLs	0.3	—
Reference exchange current density in the anode	100	A/m ³
Reference exchange current density in the cathode	0.01	A/m ³
Permeability of GDLs/CLs	$1.18 \times 10^{-11} / 2.36 \times 10^{-11}$	m ²
Electrical conductivity of the membrane	9.825	S/m
Electrical conductivity of GDLs	222	S/m
Anode/Cathode exchange coefficient	1	—

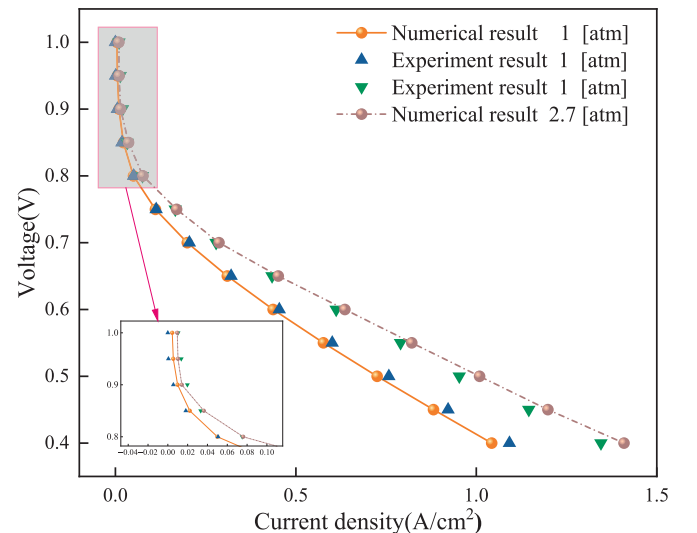


Fig. 2. Comparison of numerical simulation polarization curves with experimental results.

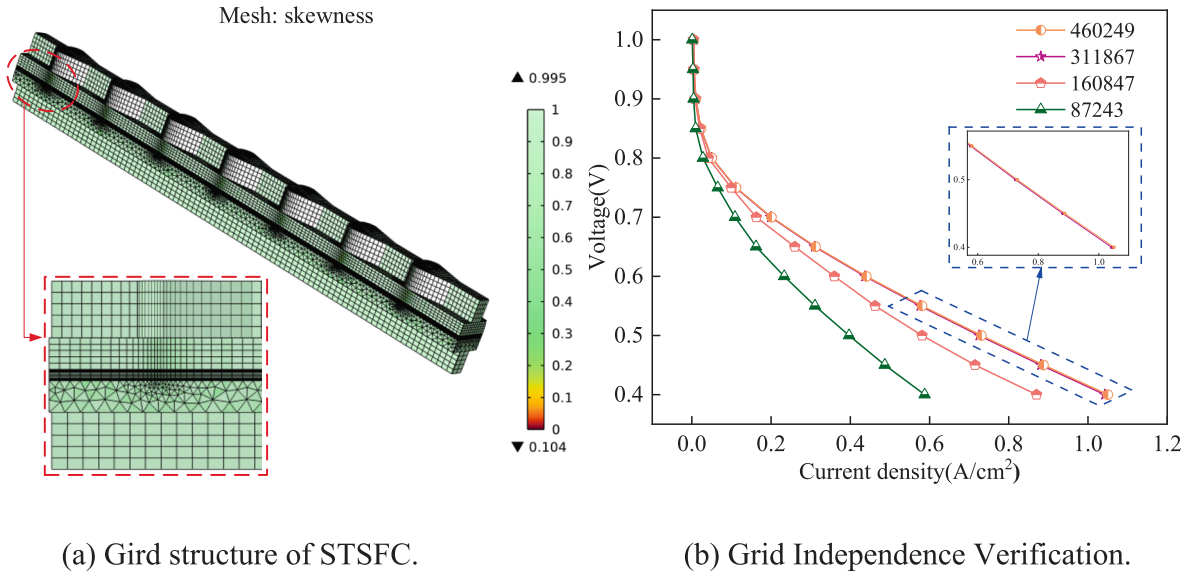


Fig. 3. The influence of grid number on calculation results.

levels of sub-goals, and specific alternatives. The priority weights of elements in each level are calculated by deriving the eigenvectors of the corresponding pairwise comparison matrices.

The first step is to divide the decision-making problem into the top level, middle level, and bottom level based on the relationships among the overall goal, decision criteria, and alternatives, and to construct the hierarchical structure accordingly, as shown in Fig. 5. For any two adjacent levels, the higher one is referred to as the target layer, and the lower one as the factor layer.

The second step involves performing pairwise comparisons among factors at each level using a relative scale, in order to reduce the difficulty of comparing heterogeneous criteria and to improve the accuracy of the judgment. Invite experts to judge the importance of each factor based on Satty's scale weight Table 5 and construct judgment matrix A:

$$A = \begin{bmatrix} 1 & a_{12} & \cdots & a_{1n} \\ a_{21} & 1 & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & 1 \end{bmatrix} \quad (12)$$

where, a_{ij} is the comparison results of the importance of factor i and factor j , and

$$a_{ij} = \frac{1}{a_{ji}} \quad (13)$$

Finally, to ensure the reliability of the results, the consistency index (CI) and consistency ratio (CR) are introduced to evaluate whether the pairwise judgment matrix satisfies consistency. The specific calculation formulas are as follows:

$$CI = \frac{\lambda_{\max} - n}{n - 1} \quad (14)$$

where, $\lambda_{\max}$ is the maximum eigenvalue of the judgment matrix, n is the total number of influencing factors.

$$CR = \frac{CI}{RI} \quad (16)$$

where, RI is the random index values, and the value of RI is determined by the n (see Table 6).

Generally, if $CR < 0.1$, the pairwise judgment matrix is considered to have acceptable consistency; otherwise, it is deemed to lack satisfactory

consistency.

2.4.2. Entropy weight method

Based on information theory and the concept of entropy, Shannon [42] developed an objective weighting method in which information entropy reflects the degree of data dispersion, thereby determining the importance of each indicator and avoiding subjective bias. According to this method, the greater the variation in an indicator, the higher its assigned weight. As a result, the Entropy Weight Method (EWM) has been widely applied in decision-making processes in recent years.

Firstly, using the normalized data from Section 2.5, a non-negative matrix for n evaluation objects and m evaluation criteria is constructed, as shown in Eq. (16). Each element P_{ij} in the probability matrix P is calculated using Eq. (17).

$$Z = \begin{bmatrix} z_{11} & z_{12} & \cdots & z_{1m} \\ z_{21} & z_{22} & \cdots & z_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ z_{n1} & z_{n2} & \cdots & z_{nm} \end{bmatrix} \quad (16)$$

Then, the information entropy of the j_{th} indicator is calculated using the following formula:

$$e_j = -\frac{1}{\ln n} \sum_{i=1}^n P_{ij} \ln(P_{ij}) \quad (i = 1, 2, \dots, n; j = 1, 2, \dots, m) \quad (17)$$

Finally, the entropy weight of each indicator can be obtained:

$$\omega_{Ej} = \frac{1 - e_j}{\sum_{j=1}^m (1 - e_j)} \quad (18)$$

2.4.3. AHP-EWM determination of comprehensive weights

Based on the combined AHP-EWM analysis method, this study derives the comprehensive weights of each factor (geometric parameters). It is concluded that the integrated model provides results that are more consistent with practical conditions than those obtained using either method alone [41]. The objective function is defined as follows:

$$\min F = \sum_{j=1}^m \omega_j \ln\left(\frac{\omega_j}{\omega_{A,j}}\right) + \sum_{j=1}^m \omega_j \ln\left(\frac{\omega_j}{\omega_{E,j}}\right) \quad (19)$$

$$s.t. \sum_{j=1}^m \omega_j = 1, \omega_j > 0 \quad (20)$$

where, $\omega_{A,j}$ is the subjective weights calculated by AHP, $\omega_{E,j}$ is the

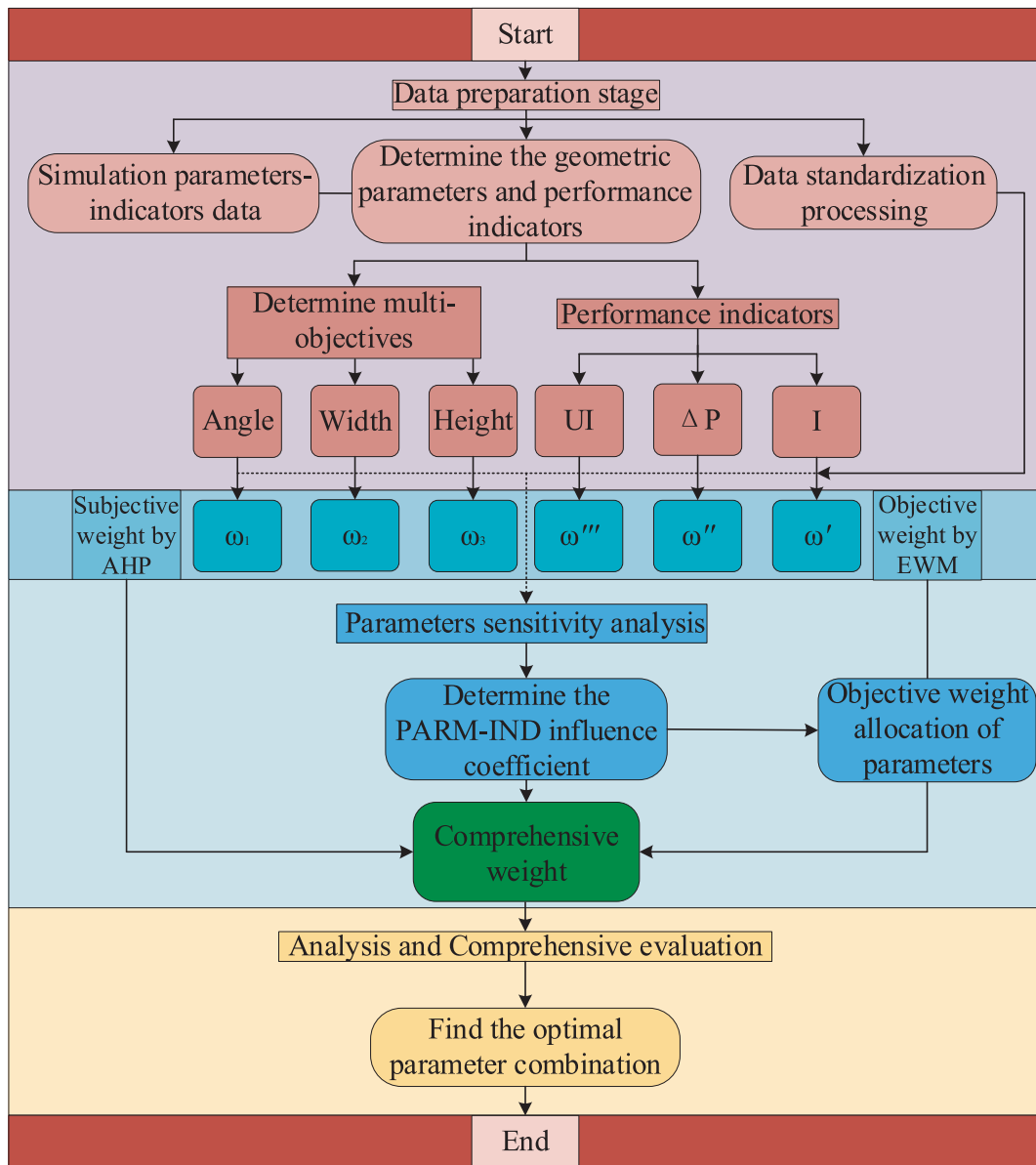


Fig. 4. Flowchart of the AHP-EWM decision framework.

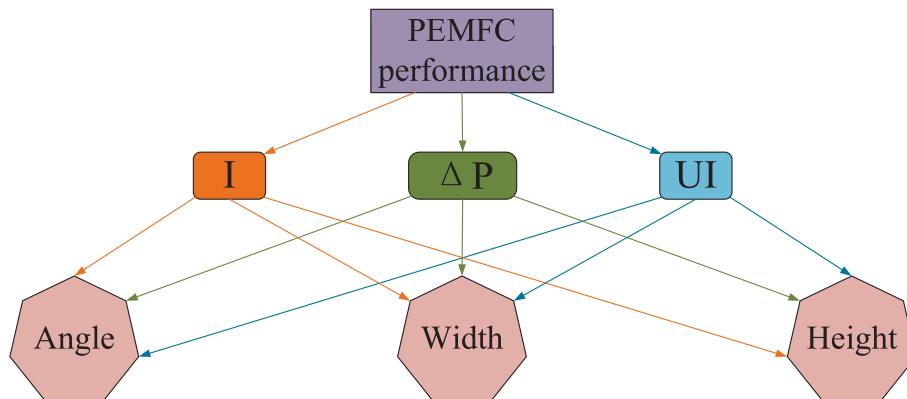


Fig. 5. Schematic diagram of structural hierarchy.

Table 5

Satty's scale weight in AHP [41].

Factor i compared with factor j	Quantitative value
Equally important	1
Moderately important	3
Strongly important	5
Very strongly important	7
Extremely important	9
The intermediate situation between the above two adjacent judgments	2, 4, 6, 8
Inverse comparison	Reciprocals

objective weights calculated by EWM. The comprehensive weights are obtained using the linear weighted aggregation method.

$$\omega_{\text{com}} = \alpha \bullet \omega_A + (1 - \alpha) \bullet \omega_E \quad (21)$$

where, α is the resolution factor, with α usually taken as 0.5 [43].

2.5. Data Standardization and evaluation indicators

Performance indicators of PEMFCs can be categorized into positive and negative indicators. For example, current density (I) is a positive indicator—higher values at a given operating voltage indicate better PEMFC performance. In contrast, pressure drop (ΔP) and oxygen uniformity index (UI) are negative indicators: a higher pressure drop indicates greater parasitic losses in the PEMFC system, and a larger oxygen uniformity index reflects poorer distribution of oxygen within the flow channels. Therefore, to eliminate dimensional inconsistencies across different indicators, all metrics are normalized to a range between 0 and 1 [44]. After normalization, the degree of variation of each indicator reflects only its intrinsic influence on the overall outcome. Positive and negative indicators are standardized by [45]:

$$Z_{ij} = \frac{X_i - \min(X_i)}{\max(X_i) - \min(X_i)} \quad (22)$$

$$Z_{ij} = \frac{\max(X_i) - X_i}{\max(X_i) - \min(X_i)} \quad (23)$$

where, Z_{ij} is the standardized value, X_i is the original value of the j_{th} factor, $\max(X_i)$ is the maximum of the j_{th} factor, $\min(X_i)$ is the minimum values of the j_{th} factor.

This study selects three evaluation objectives that influence PEMFC performance: I , ΔP , and UI at the GDL–CL interface. The latter two objectives are obtained using Eq. (24) and Eq. (25).

The ΔP is determined by the pressure difference between the inlet and the outlet:

$$\Delta P = P_{\text{in}} - P_{\text{out}} \quad (24)$$

The UI value at the interface between GDL and CL can be obtained by the following formula:

$$UI = \frac{\int_A |X - X_{\text{av}}| dA}{\int_A X_{\text{av}} dA} \quad (25)$$

where, X is the molar concentration value of oxygen, and X_{av} is the average molar concentration value in the flow channel.

3. Results and discussion

This chapter will conduct a systematic analysis of the performance of STSFC. By combining numerical simulation with multi-criteria evaluation, it will present the influence laws of key parameters on battery performance and screen out the optimal solution.

The θ , H , and W have significant effects on PEMFC performance. To systematically investigate the performance of different design combinations, we refer to the research in references [15–17] and select θ values of 30° , 60° , 90° and 120° , which can cover the effective unidirectional flow interval of the Tesla-like valve flow channel. Based on the typical dimensions of the commercial PEMFC flow field and referring to [28], the ranges of W and H are determined. Through pre-simulation verification, this range can prevent blockage caused by too narrow flow channels or decrease in mass transfer efficiency caused by too wide ones. To avoid overfitting and ensure full coverage of the parameter space, the corresponding simulation data are presented in Table 7. Fig. 6 illustrates the relationship between I , ΔP , and UI for each configuration at a working voltage of 0.45 V. The result shows that both increasing W while keeping H constant, and increasing H while keeping W constant, result in a decrease in both voltage drop and oxygen uniformity index. However, as shown in Fig. 6 (a), when H remains constant and W increases, the current density decreases. Conversely, when W remains constant and H increases, the current density increases. To explore the

Table 7

Experimental Data.

Scheme	θ	H	W	I	ΔP	UI
1	30°	0.8	0.6	0.90651	33.501	0.09087
2	30°	0.8	0.8	0.90559	25.545	0.07356
3	30°	0.8	1	0.90042	21.586	0.06399
4	30°	1	0.6	0.90863	29.709	0.07706
5	30°	1	0.8	0.90733	21.16	0.06234
6	30°	1	1	0.90175	17.034	0.05474
7	30°	1.2	0.6	0.90946	27.534	0.03985
8	30°	1.2	0.8	0.90776	18.674	0.05718
9	30°	1.2	1	0.90221	14.518	0.05069
10	60°	0.8	0.6	0.89654	32.602	0.08660
11	60°	0.8	0.8	0.89361	25.068	0.07471
12	60°	0.8	1	0.88902	21.218	0.05968
13	60°	1	0.6	0.89889	28.235	0.07252
14	60°	1	0.8	0.89497	20.731	0.06205
15	60°	1	1	0.8904	16.819	0.05085
16	60°	1.2	0.6	0.89987	26.332	0.06511
17	60°	1.2	0.8	0.89558	18.424	0.05324
18	60°	1.2	1	0.89085	14.347	0.04052
19	90°	0.8	0.6	0.89785	32.412	0.08478
20	90°	0.8	0.8	0.89415	24.948	0.06810
21	90°	0.8	1	0.8897	21.216	0.05871
22	90°	1	0.6	0.90035	28.302	0.07055
23	90°	1	0.8	0.89604	20.706	0.05740
24	90°	1	1	0.89121	16.86	0.04985
25	90°	1.2	0.6	0.90139	26.45	0.06346
26	90°	1.2	0.8	0.89673	18.484	0.05198
27	90°	1.2	1	0.89167	14.419	0.04547
28	120°	0.8	0.6	0.89813	32.461	0.08455
29	120°	0.8	0.8	0.8944	24.89	0.06777
30	120°	0.8	1	0.88989	21.168	0.05847
31	120°	1	0.6	0.90066	28.227	0.07020
32	120°	1	0.8	0.89629	20.704	0.05717
33	120°	1	1	0.89136	16.829	0.04972
34	120°	1.2	0.6	0.90171	26.293	0.06297
35	120°	1.2	0.8	0.89702	18.501	0.05177
36	120°	1.2	1	0.89185	14.39	0.04522

Table 6

RI index value table.

n	1	2	3	4	5	6	7	8	9	10
RI	0	0	0.52	0.89	1.12	1.26	1.36	1.41	1.46	1.49

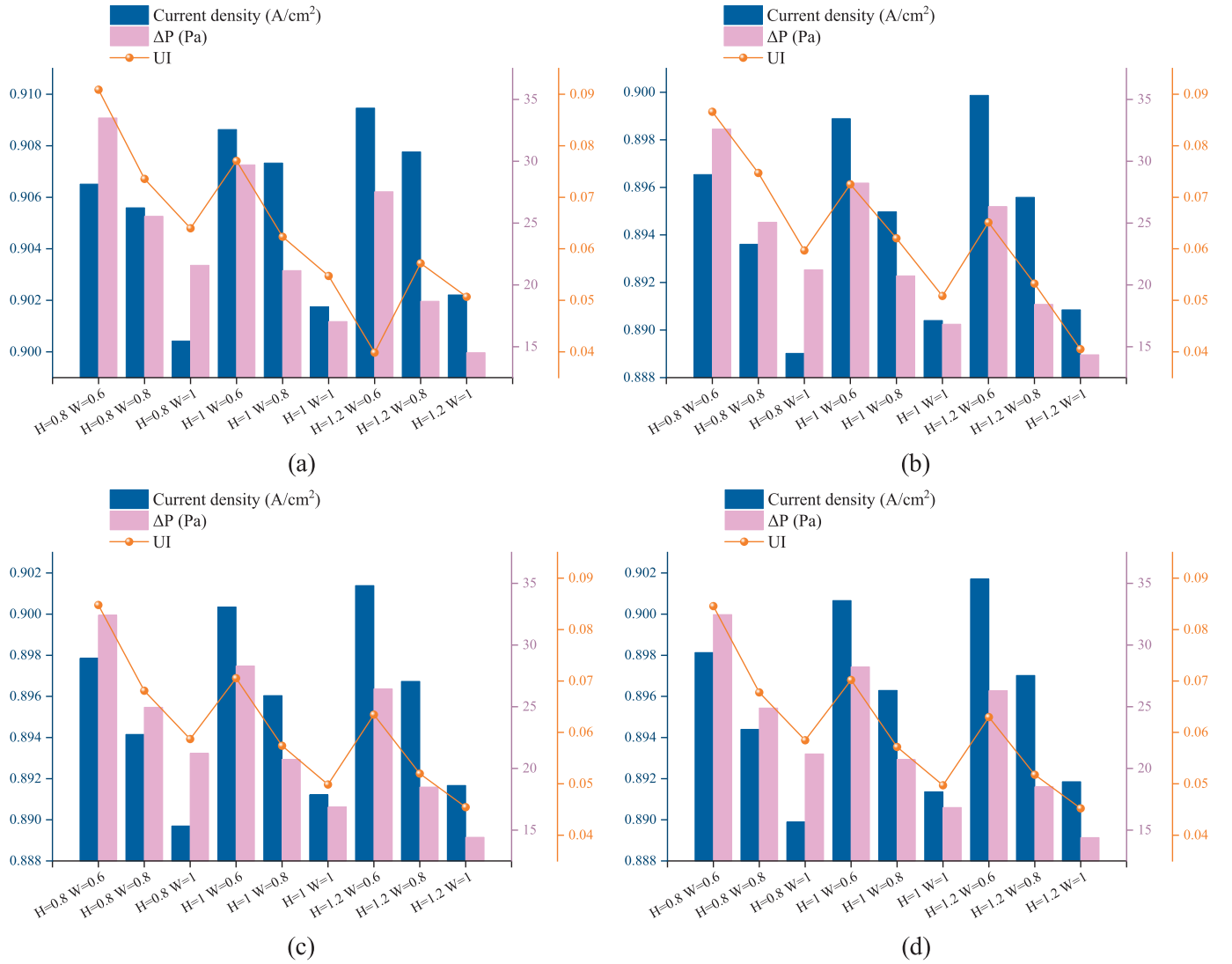


Fig. 6. Schematic diagram of I, ΔP and UI at loop angles (θ) of 30° , 60° , 90° and 120° respectively under a working voltage of 0.45 V: (a) $\theta = 30^\circ$, (b) $\theta = 60^\circ$, (c) $\theta = 90^\circ$, (d) $\theta = 120^\circ$. The data is derived from the simulation results in Table 7.

relationship between the geometric parameters and the three factors, this study will conduct further analysis to identify the optimal parameter combination for this PEMFC. For multi-criteria PEMFC performance evaluation, both subjective and objective weights should be considered. In this study, the AHP and EWM are integrated to obtain the comprehensive weight of each factor.

3.1. Weights calculation using AHP

The expert judgment results based on the importance table are as follows. It is worth noting that a total of five experts participated in the judgment, all of whom hold doctoral degrees in PEMFC flow field design or multi-criteria decision-making and have over five years of research experience (three of whom are from the energy engineering departments of universities and two from the R&D departments of new energy

enterprises). The expert judgment adopts the “independent scoring + group discussion” model: First, each expert independently fills in the judgment matrix, and then through group discussion (a total of 2 rounds), the indicators with significant scoring differences (deviation > 20 %) are resolved. Finally, a unified judgment matrix (Table 8-11) is formed to ensure the objectivity and consistency of the results.

The above results are organized into a judgment matrix as follows:

$$A_{O-C} = \begin{bmatrix} 1 & 7 & 4 \\ \frac{1}{7} & 1 & \frac{1}{3} \\ \frac{1}{4} & 3 & 1 \end{bmatrix} \quad (26)$$

Table 8
Expert judgment results O-C.

Objectives	I	ΔP	UI
I	1	7	4
ΔP	1/7	1	1/3
UI	1/4	3	1

Table 9
C-P_i.

I	θ	H	W
θ	1	1/2	1/5
H	2	1	1/3
W	5	3	1

Table 10C-P_{ΔP}.

ΔP	θ	H	W
θ	1	1/5	2
H	5	1	7
W	1/2	1/7	1

Table 11C-P_{UI}.

UI	θ	H	W
θ	1	5	1/3
H	1/5	1	1/5
W	3	5	1

$$A_{C-P_1} = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{5} \\ 2 & 1 & \frac{1}{3} \\ 5 & 3 & 1 \end{bmatrix} \quad (27)$$

$$A_{C-P_{\Delta P}} = \begin{bmatrix} 1 & \frac{1}{5} & 2 \\ 5 & 1 & 7 \\ \frac{1}{2} & \frac{1}{7} & 1 \end{bmatrix} \quad (28)$$

$$A_{C-P_{UI}} = \begin{bmatrix} 1 & 3 & \frac{1}{2} \\ \frac{1}{3} & 1 & \frac{1}{3} \\ 2 & 3 & 1 \end{bmatrix} \quad (29)$$

Let the maximum eigenvalue and the normalized weight of the above matrix be $\lambda_{max1}, \lambda_{max2}, \lambda_{max3}, \lambda_{max4}$ and $\Omega_1, \Omega_2, \Omega_3, \Omega_4$, respectively. The calculated maximum eigenvalues are $\lambda_{max1} = 3.0536$, $\lambda_{max2} = 3.0092$, $\lambda_{max3} = 3.0897$, $\lambda_{max4} = 3.0037$, with the corresponding normalized eigenvector weights are $\Omega_1 = (0.7050 \ 0.0832 \ 0.2118)^T$, $\Omega_2 = (0.1000 \ 0.2400 \ 0.6600)^T$, $\Omega_3 = (0.2100 \ 0.6200 \ 0.1700)^T$, $\Omega_4 = (0.2715 \ 0.1145 \ 0.6140)^T$.

Consistency verification is conducted using Eq. (30) and Eq. (31):

$$CI_i = \frac{\lambda_{max_i} - n}{n - 1} \quad (30)$$

$$CR_i = \frac{CI_i}{RI} \quad (31)$$

The calculated values are: $CI_1 = 0.0268$, $CI_2 = 0.046$, $CI_3 = 0.0449$ and $CI_4 = 0.0019$. According to the RI table, the corresponding RI value is 0.52. Therefore, the consistency ratios are $CR_1 = 0.0515$, $CR_2 = 0.0885$, $CR_3 = 0.0863$ and $CR_4 = 0.0037$. Since all values are less

Table 12

Weights of assessment factors calibrated from AHP.

Index layer Index	AHP weight	Sub-index layer Factor	AHP weight
I	0.71	θ	0.10
		H	0.24
		W	0.66
ΔP	0.08	θ	0.21
		H	0.62
		W	0.17
UI	0.21	θ	0.27
		H	0.12
		W	0.61

than 0.1, the consistency is acceptable and the weights are considered valid.

The above results are summarized in Table 12. Subsequently, the subjective weights of the geometric parameters affecting PEMFC performance are calculated using the arithmetic mean method, and the results are as follows.

$$\omega_{A,\theta} = 0.19$$

$$\omega_{A,H} = 0.33$$

$$\omega_{A,W} = 0.48$$

As shown in Table 12, the weights of each element with respect to its upper-level element are summarized. The results indicate that the weights assigned to I, ΔP, and UI are 0.71, 0.08, and 0.21, respectively. Subjective evaluation suggests that current density is the most important performance indicator for PEMFCs. ΔP, with the lowest weight, is considered the least important performance metric. At the lowest (factor) level, the weights of W with respect to I and UI are 0.66 and 0.61, respectively—both higher than the weights of θ and H with respect to I. In terms of ΔP, H has the highest weight at 0.62.

3.2. Weights calculation using EWM

According to Table 7, the 36 data sets are normalized to values between 0 and 1 to obtain the standardized matrix Z. The results are shown below.

$$Z = \begin{bmatrix} 0.8857 & 0 & 0 \\ 0.8107 & 0.4154 & 0.3393 \\ \vdots & \vdots & \vdots \\ 0.3914 & 0.7831 & 0.7664 \\ 0.1385 & 0.9978 & 1 \end{bmatrix} \quad (32)$$

After data normalization, the weight of the i_{th} factor under the j_{th} indicator is calculated using Eq. (33), and the results are as follows:

$$P_{ij} = \begin{bmatrix} 0.05551 & 0 & 0 \\ 0.05259 & 0.02063 & 0.01668 \\ \vdots & \vdots & \vdots \\ 0.02539 & 0.03889 & 0.03768 \\ 0.00898 & 0.04954 & 0.04399 \end{bmatrix} \quad (33)$$

Based on Eq. (17) and Eq. (18), the information entropy e_j and the entropy weight $\omega_{E,j}$ are calculated. The results are shown in Table 13.

To obtain the specific entropy weight of each geometric parameter for the PEMFC, it is necessary to calculate the influence coefficient of each parameter on the performance indicators. Sensitivity analysis provides a direct measure of the impact intensity of geometric parameters on PEMFC performance. Therefore, by integrating the entropy weight method with sensitivity analysis, the rationality of the weight assignment is optimized from both data-driven and physical perspectives, thereby providing stronger support for the analysis and conclusions of this study. The analysis process is as follows.

Based on the data in Table 7, this study adopts a One-At-a-Time (OFAT) Sensitivity Analysis, using ($\theta = 60^\circ$, $H = 1$, $W = 0.8$) as the baseline configuration. The relative sensitivity coefficients of each

Table 13the values of e_j and $\omega_{E,j}$.

	I	ΔP	UI
e_j	0.9326	0.9526	0.9661
$\omega_{E,j}$	0.45	0.32	0.23

parameter with respect to I, ΔP , and UI are calculated using Eq. (34). These sensitivity coefficients are then used to determine the objective weights of the geometric parameters- $\omega_{E,GRKSLT}$, $\omega_{E,H}$ and $\omega_{E,W}$ -in relation to PEMFC performance indicators. Eq. (34) is given as follows:

$$S = \frac{\Delta Y/Y_0}{\Delta X/X_0} \quad (34)$$

where, Y represents the performance indicator, X denotes the geometric parameter, and subscript 0 refers to the baseline value. The results are shown in Table 14.

According to the research of Loksupapaiboon et al. [46], a heat map of the Pearson correlation coefficient between the input parameters and the output sensitivity is presented, as shown in Fig. 7. For I, the relative sensitivity coefficient of H (0.035) is slightly higher than that of W (0.02) and θ (0.028), and all three show a very weak correlation. For ΔP , the relative sensitivity coefficient is the highest for W (1.448), followed by H (1.045), and finally θ (0.041). It can be seen that ΔP shows a strong correlation with W and H. For UI, the relative sensitivity coefficients of H and W are 0.722 and 0.71 respectively, showing the same correlation. The correlation of θ is the lowest, with a relative sensitivity coefficient of 0.15.

The above results are summarized in Table 15, where the influence coefficients derived from sensitivity analysis are integrated with the EWM results as part of the objective weighting scheme. The resulting objective weights of the geometric parameters in relation to PEMFC performance are as follows.

$$\omega_{E,\theta} = 0.34 \times 0.45 + 0.02 \times 0.32 + 0.09 \times 0.23 = 0.18$$

$$\omega_{E,H} = 0.42$$

$$\omega_{E,W} = 0.40$$

Table 15 reveals the relative importance of the performance indicators for PEMFC: current density I dominates with an objective weight of 0.45, followed by pressure drop ΔP (0.32) and the oxygen uniformity index UI (0.23). This aligns with the practical operational goal of maximizing output current in PEMFCs, while also acknowledging the significant roles of ΔP and oxygen concentration uniformity in affecting performance. The results, after incorporating sensitivity analysis, show that variations in flow channel dimensions have a more pronounced impact on PEMFC performance than changes in θ . Specifically, W has the greatest influence on ΔP (sensitivity coefficient = 1.448) and UI (0.722), primarily by regulating flow resistance and reactant distribution within the channels. H plays a dominant role in UI (sensitivity = 0.71) and significantly affects ΔP (1.045) by altering the thickness of the diffusion path and thus influencing mass transfer efficiency. In contrast, θ plays a secondary role, with its highest sensitivity observed for UI (0.15). This indicates that under the operating conditions of this study, variations in θ have some influence on oxygen distribution uniformity, but to a lesser extent than H and W.

Table 14
Results of sensitivity analysis.

Performance indicators	Parameter	Relative sensitivity coefficient (S)	Normalized influence coefficient (α_i)
I	θ	0.028	0.34
	H	0.035	0.42
	W	0.020	0.24
ΔP	θ	0.041	0.02
	H	1.045	0.41
	W	1.448	0.57
UI	θ	0.15	0.09
	H	0.71	0.45
	W	0.722	0.46

3.3. AHP-EWM determination of comprehensive weights

Fig. 8 presents the comprehensive weights of each performance indicator and factor as determined by the combined AHP-EWM analysis method. Among the primary indicators, I has the highest comprehensive weight at 0.58, followed by the oxygen uniformity index UI at 0.22, and ΔP with the lowest at 0.20. For the secondary indicators, the W has the highest weight at 0.44, followed by H (0.375) and θ (0.185). The integration of the two methods reveals that the importance of geometric parameters varies depending on the performance indicator. For instance, the weight of W is highest for both I (0.45) and UI (0.535). In contrast, channel height H holds the highest weight for ΔP , at 0.515. The weight of θ remains relatively stable across all three performance indicators. These results suggest that in the STSFC geometry, channel height and width are the two most critical parameters to consider for performance optimization. It is worth noting that this study focuses solely on the θ of the Tesla structure channel (TSC), without evaluating other design aspects of the TSC. Future work could adopt more comprehensive evaluation methods to explore additional geometric parameters of the TSC design.

The Coefficient of Variation (CV) is a standardized indicator for measuring the degree of data dispersion and can eliminate the influence of dimensions. To ensure the accuracy of the AHP-EWM model, the coefficient of variation is introduced to compare the consistency of different weights. Its calculation formula is:

$$CV = \frac{\tau}{\bar{\mu}} \times 100\% \quad (35)$$

$$\tau = \sqrt{\frac{\sum_{i=1}^n (W_i - \bar{\mu})^2}{n}} \quad (36)$$

$$\bar{\mu} = \frac{W_A + W_E + W_{com}}{n} \quad (37)$$

where, τ is the standard deviation of a set of weighted data, $\bar{\mu}$ is the average value of the weighted data in this group. The final result is expressed as a percentage. The smaller the value, the lower the degree of data dispersion and the better the consistency of the weights (it is generally believed that when $CV < 10\%$, the consistency meets the requirements of engineering analysis).

Through calculation, it can be obtained that $CV_\theta = 2.7\%$, $CV_W = 9.09\%$, $CV_H = 12\%$. (It is worth noting that the weight difference of H stems from the slight deviation between the “influence of height on pressure drop” in the subjective judgment of AHP and the “influence of height on oxygen uniformity” in the objective data of EWM. However, the comprehensive weight has balanced this difference through linear fusion, and 12% is still within the small discrete range acceptable to engineering, which does not affect the reliability of the overall weight). The average coefficient of variation of the three groups of parameters is 7.93%, and the overall figure is less than 10%, indicating that the consistency between subjective and objective weights is good, and the comprehensive weights could reliably reflect the actual influence degree of geometric parameters on the performance of PEMFC.

3.4. Comprehensive analysis

After determining the final weights, the overall score of each alternative can be calculated using Eq. (36).

$$S_{com} = \omega_{comI} \times I_s + \omega_{com\Delta P} \times \Delta P_s + \omega_{comUI} \times UI_s \quad (38)$$

where, ω_{com} is the comprehensive weights of each first-level indicator, I_s , ΔP_s and UI_s represent the standard values of I, ΔP and UI respectively. The calculation results are shown in Fig. 9.

As shown in Fig. 9, Scheme 7 achieves the highest overall score of 0.8623. The geometric parameters of Scheme 7 include a θ of 30° , W of

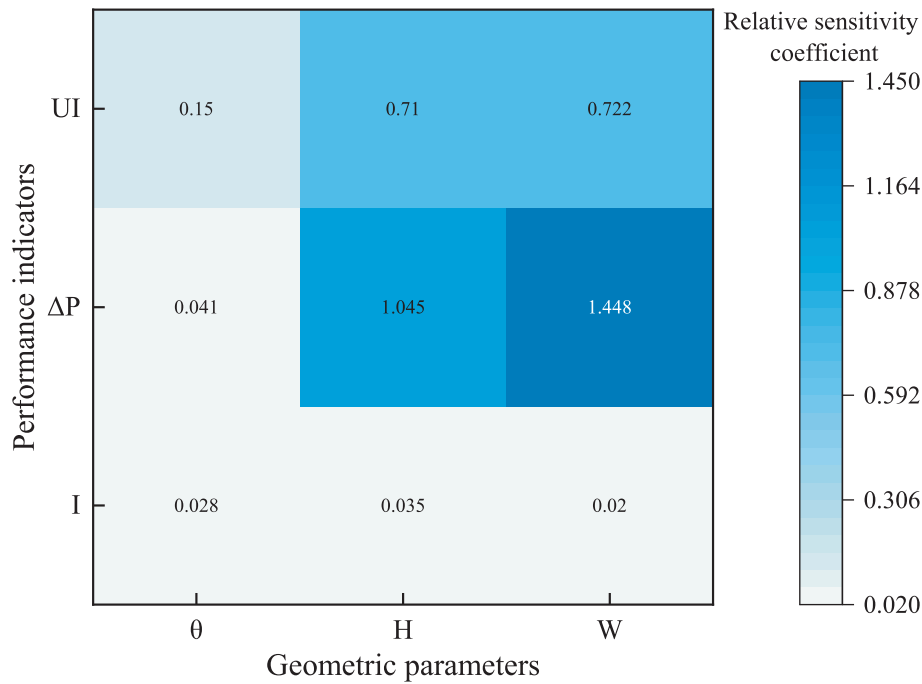


Fig. 7. Heatmaps of Pearson correlation coefficients for input parameters and the output sensitivity. The data are derived from the sensitivity analysis results in Table 14.

Table 15
Weights of assessment factors calibrated from EWM.

Index layer	EWM weight	Sub-index layer	EWM weight
Index		Factor	
I	0.45	θ	0.34
		H	0.42
		W	0.24
ΔP	0.32	θ	0.02
		H	0.41
		W	0.57
UI	0.23	θ	0.09
		H	0.45
		W	0.46

0.6 mm, and H of 1.2 mm. It also exhibits the highest I (0.9049 A/cm²) and the best UI (0.03985) among all cases. The ΔP, however, is moderate at 27.534 Pa. These results suggest that, in multi-objective optimization of PEMFCs, focusing on the dominant performance metrics can be sufficient for identifying the optimal design configuration.

Fig. 10 shows the changing trends of I, ΔP, and UI, based on Scheme 7 (θ = 30°, H = 1.2, W = 0.6), when only the θ, H, and W are changed. Fig. 10 (a) shows that when the θ increases, the intensity of the secondary flow in the flow channel weakens, which cannot effectively promote the transfer of O₂ to the CL, causing UI to increase from 0.03985 to around 0.06. However, increasing the θ will reduce the flow resistance, with ΔP decreasing from 27.53 Pa to around 26.3 Pa. I is less affected by θ (change rate < 2%), as the current density mainly depends on the total supply of reactants rather than the local flow field structure. It can be seen from Fig. 10 (b) that when H increases, the cross-sectional area of the flow channel increases, and ΔP drops from 33.50 Pa to 27.53 Pa. Meanwhile, the increase of H increases the contact area between GDL and the flow channel, shortens the oxygen transport path, and reduces UI from 0.090 to 0.039. I slightly increased with the increase of H (from 0.906 A/cm² to 0.909 A/cm²), as the reduced mass transfer resistance promoted the reaction. Fig. 10 (c) shows that when W increases, the flow velocity in the channel decreases, weakening the convective mass, and UI increases accordingly. Similarly, I slightly

decreases with the increase of W (from 0.909 A/cm² to 0.902 A/cm²). However, an increase in W will cause ΔP to decrease from 27.53 Pa to 14.52 Pa.

Fig. 11 compares the performance of the optimal parameter combination, Scheme 7, with that of the conventional parallel flow channel (CPFC). As shown in Fig. 11 (a), the current density of both configurations is nearly identical before reaching a working voltage of 0.6 V. However, as the voltage decreases and the system enters the high current density region, the PEMFC reaction rate is primarily controlled by concentration polarization, leading to a widening gap between the two configurations. At the same voltage, STSFC achieves a current density of 1.9436 A/cm², which is 12.3 % higher than that of CPFC. The maximum peak power density reaches 0.43803 W/cm², which is 17.4 % higher than that of CPFC. Fig. 11 (b and c) clearly show that STSFC outperforms CPFC in terms of gas pressure loss and oxygen concentration uniformity. These results support the potential of STSFC in reducing system energy consumption and enhancing fuel supply capacity.

4. Conclusions

This study proposes a multi-objective optimization and evaluation of STSFC performance using a combined AHP–EWM analysis method to assess its flow field structural parameters. The relative importance of TSC θ, H, and W to STSFC performance is analyzed, including their individual contributions to I, ΔP, and UI. The main conclusions are as follows:

- (1) For overall STSFC performance, the importance ranking of the performance indicators is: I > UI > ΔP, with weights of 0.58, 0.22, and 0.20, respectively. For geometric parameters, the order of importance is: W > H > θ, with weights of 0.44, 0.375, and 0.185, respectively. This indicates that current density plays a dominant role in performance evaluation, while channel width has the greatest impact on overall STSFC performance.
- (2) According to the OFAT, W and H exhibit significantly higher sensitivity toward ΔP and UI than θ; I shows relatively low sensitivity to all parameters.

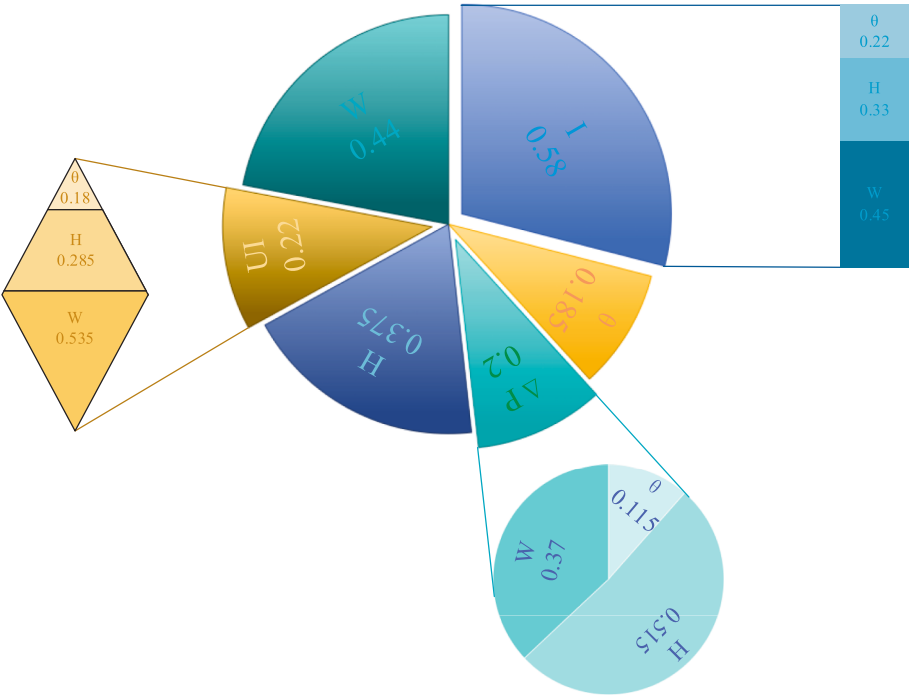


Fig. 8. Weight distribution of each indicator.

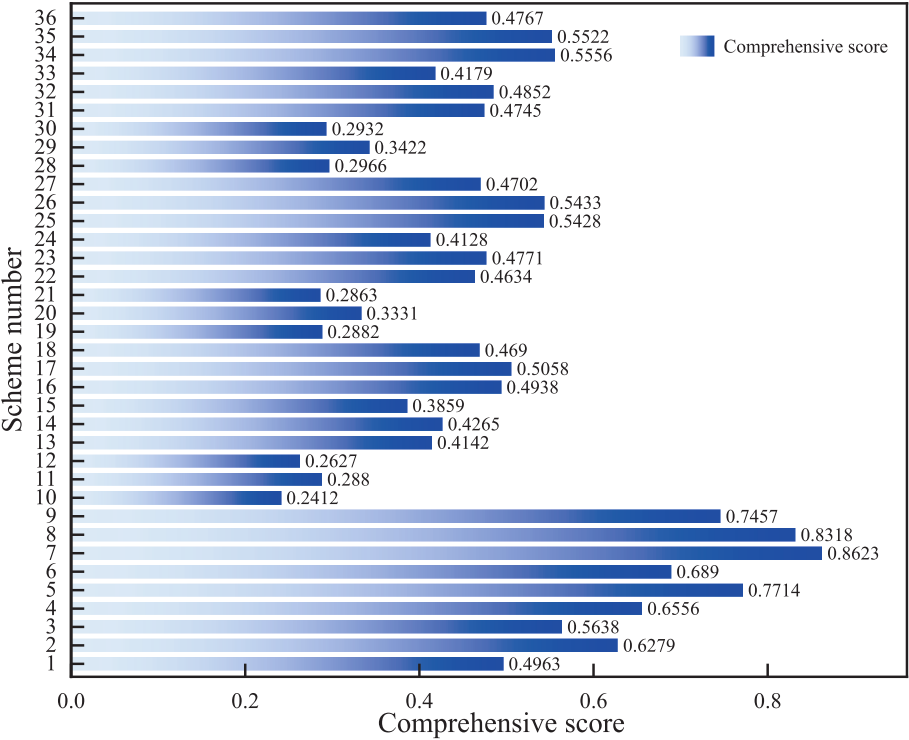


Fig. 9. The trend of AHP-EWM comprehensive score.

- (3) According to the final ranking based on the comprehensive scores, Scheme 7 achieves the highest score of 0.8623. The optimal geometric parameter combination is $\theta = 30^\circ$, $H = 1.2$ mm, and $W = 0.6$ mm. This configuration offers the best balance among current density, pressure drop, and oxygen distribution uniformity in this study.
- (4) During the process of optimizing the performance of PEMFC this time, it is found that the sensitivity of W and H to ΔP is 1.448 and 1.045, respectively, and the sensitivity to UI is 0.722 and 0.71, respectively, which is significantly higher than that of the θ at 0.041 and 0.15. This provides engineering guidance for the optimization of the PEMFC flow field, which is “prioritize

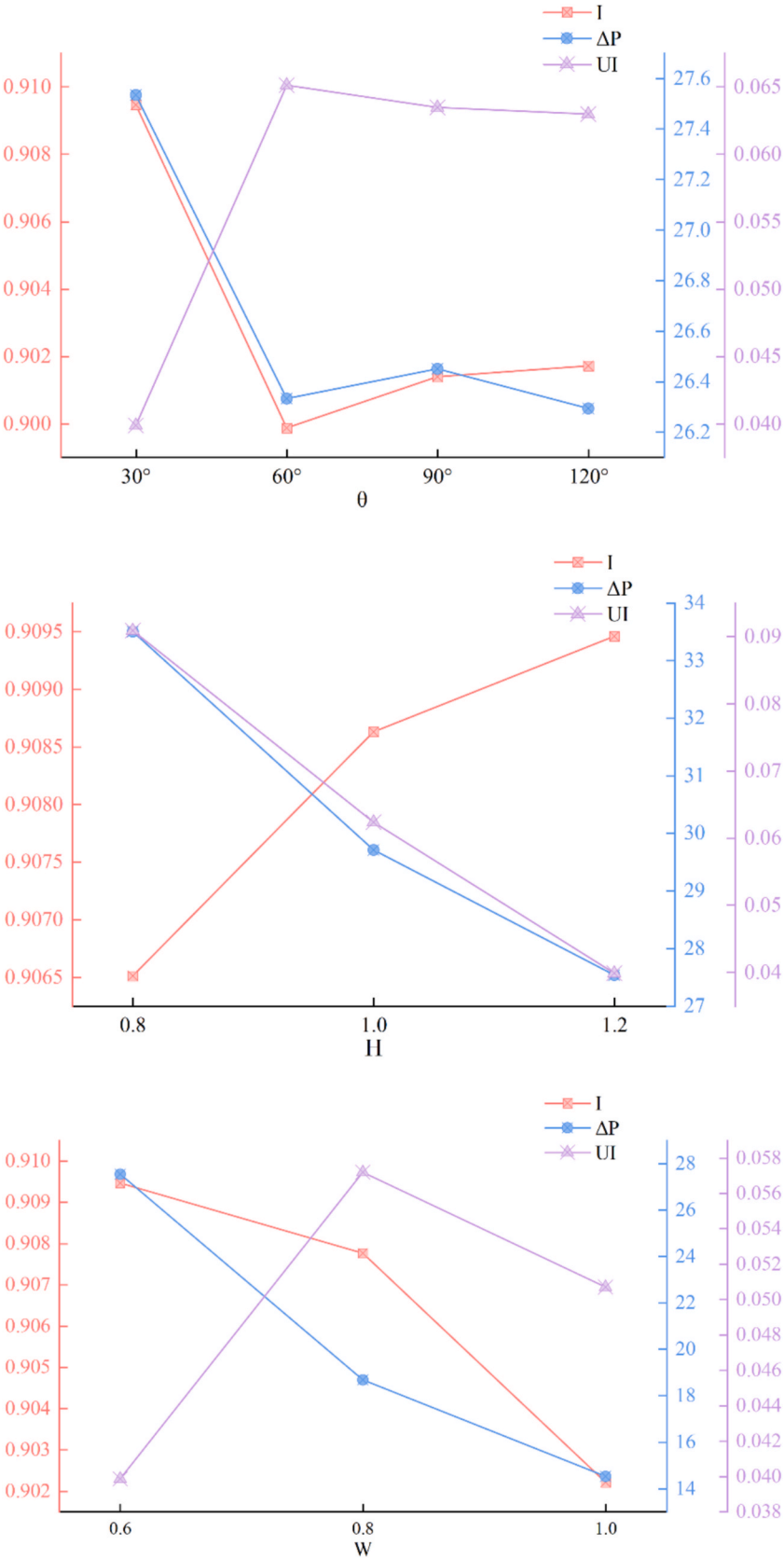


Fig. 10. Trend chart of the influence of Tesla valve parameters on the performance of PEMFC. (a) θ -I/ ΔP /UI trend, (b) H-I/ ΔP /UI trend, (c) W-I/ ΔP /UI trend. The data is derived from the simulation results in Table 5.

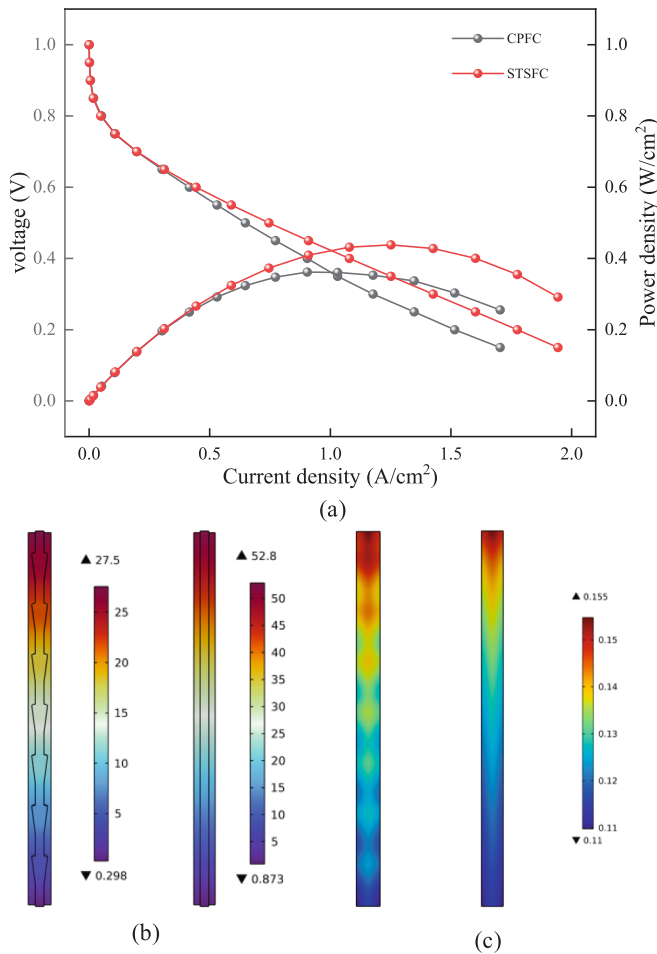


Fig. 11. Simulation performance comparison between STSFC with the optimal parameter combination and conventional parallel flow channel (CPFC): (a) Polarization curve and power density curve, (b) pressure drop distribution diagram, (c) Oxygen concentration distribution diagram at the interface between GDL and CL.

adjusting the W and H , and then optimize the θ ", rather than the traditional "full-parameter equivalent optimization".

- (5) In the process of optimizing PEMFC performance, it is sufficient to identify and further refine the top two or three parameters that have the greatest impact using this method. In this study, we focus on the impact of changes in a single geometric structure on the important performance indicators of the battery, and thus employ the OFAT method. However, using this method is prone to missing parameter interactions. In the future, the response surface method (RSM) will be adopted to further analyze parameter interactions and optimize the design space.

CRedit authorship contribution statement

Wenxie Bu: Writing – original draft, Validation, Software, Methodology. **Han Liang:** Writing – original draft, Visualization, Validation, Software. **Wang Gu:** Writing – original draft, Visualization, Investigation, Data curation. **Minggang Zheng:** Writing – review & editing, Supervision, Resources, Project administration, Conceptualization. **Yitzhak Cohen:** Resources, Project administration, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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